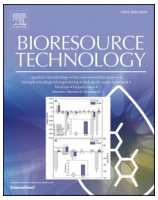
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Micro-aeration strategy for improved biogas yield and in-situ hydrogen sulfide mitigation during high-solids co-digestion of poultry litter

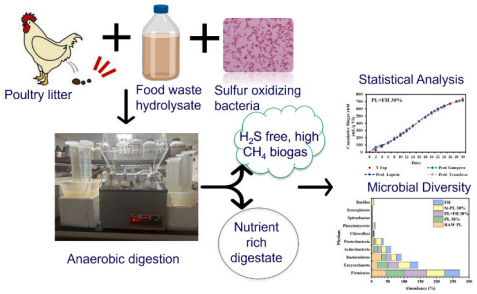
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HIGHLIGHTS

- Higher solid concentrations (up to 30% Total Solids) improved biogas production.
- Micro-aeration improved biogas yield and reduced H_2S emissions.
- Carbon/Nitrogen balancing enhanced process stability and methane rich biogas output.
- Metagenomics revealed microbial shifts with solids concentration changes.

GRAPHICAL ABSTRACT



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ABSTRACT

High-solids anaerobic digestion sustainably converts organic waste into methane-rich biogas and nutrient-rich biomanure, offering dual benefits. This study assessed poultry litter (PL) digestion at 10 %, 30 %, and 60 % total solids (TS) under three setups: (i) PL alone, (ii) co-digestion with food waste hydrolysate (PL + FH) to improve carbon/nitrogen (C/N) ratio, and (iii) in situ desulfurization (St-PL) using sulfur-oxidizing bacteria (SOB) with micro-aeration to reduce H_2S. The highest biogas yields were obtained at 30 % TS: 12.4 L (PL + FH), 9.9 L (PL), and 8.9 L (St-PL). Methane content was highest in St-PL (68 %) with H_2S removal of 80–92 %. However, slight sulfate accumulation was detected in the digestate, attributed to the metabolic activity of sulfur oxidizing bacteria (SOB). Microbial analysis revealed the dominance of *Firmicutes*, *Bacteroidetes, Methanosaeta, and Methanobacterium*. This study demonstrates a scalable strategy in which micro-aeration combined with SOB enhanced methane yield and biogas quality while effectively mitigating H2S during high-solids poultry litter digestion.

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1. Introduction

The accumulation of poultry litter, a major by-product of the poultry industry, poses significant environmental challenges due to its high nitrogen content which leads to ammonia emissions, nutrient leaching and groundwater pollution, the release of greenhouse gases (GHGs), and the spread of pathogens. It is estimated that in 2024, the global chicken meat production was ~103,046 tons, which is associated with the generation of approximately 68 billion tons of poultry waste annually (Coêlho et al., 2025). Over the past five decades, the production of poultry litter (PL) has increased by an average of 5 % per annum, along with other associated wastes, including droppings; bedding materials such as straw, sawdust, wood shavings, and rice hulls; hatchery waste; feathers; and processing wastewater generated during and after the production period (Beausang et al., 2020; Moreki & Keaikitse, 2013; Mottet & Tempio, 2017).

These waste materials possess properties that can substantially exacerbate pollution, leading to health hazards. Current treatment methods for poultry litter include biological, physico-chemical and thermal methods. However, most of these approaches are not cost-effective and are complex to implement, whereas biological treatment through the anaerobic digestion (AD) approach offers the dual benefit of cost-effectiveness and the generation of biogas and biomanure, contributing to the circular bioeconomy. AD is widely utilised for the treatment of a diversity of organic wastes, including cattle and pig manure, which share similar characteristics with poultry litter, particularly the hazardous emissions of H_2S and ammonia (NH_3). These wastes cause contamination through pathogens, odours, and pests, while manure facilities emit CO_2, CH_4, and N_2O, contributing to global warming. The sulfur content in poultry litter hinders the metabolism and growth of methanogenic archaea by promoting toxic sulfide production and by allowing sulfur-reducing bacteria (SRB) to compete for essential substrates. Poultry farms supplement feeds with sulfur-containing amino acids like methionine and cysteine, as chicks cannot synthesize them; these are essential for protein synthesis, feather growth, and improved egg production (Capson-Tojo et al., 2017). PL, rich in sulfur amino acids and organics, generates sulfur by-products during AD, creating microbial competition that lowers biogas yield and methane production (Mahdy et al., 2020).

Based on the total solids (TS) content of the feedstock, AD can be categorised into three types; wet AD, semi-dry AD, and dry AD or solid-state AD (Arelli et al., 2021). Compared to wet AD (TS 10 %), semi-dry AD (15–20 %), dry or solid-state AD (TS > 20 %) has recently gained significant research attention due to its various benefits including reduced water demand, higher methane yield, and less digestate generation. Although AD is typically divided into four phases, all reactions occur simultaneously and are highly interdependent. Each phase is associated with a distinct microbial community with diverse microbial consortia. The solid-state anaerobic digestion (SSAD) method was initially employed for treating the organic fraction of municipal solid waste (Li et al., 2011; Begum et al., 2024). However, it has recently been proposed for the treatment of agricultural crop residues, livestock, and poultry waste, often in combination with energy crops, to produce a stable digestate in solid phase (Chiumenti et al., 2018). Despite its promise, SSAD technology requires further development and investigation to reach maturity and expand the range of substrates it can effectively process.

The digestibility of carbon and nitrogen sources is critical in determining the optimum C/N ratio for AD. PL typically has low C/N ratio, ranging from 7 to 10, which is suboptimal for the AD process. However, Kirchmann & Witter (1992) observed that the C/N ratio increases during the AD of PL in uncovered containers, likely due to preferential loss of carbon (as CO_2) relative to nitrogen under those conditions. Understanding the roles of anaerobic bacteria such as hydrolytic, acidogenic, acetogenic and syntrophic bacteria and methanogenic archaea is critical for monitoring the AD process, avoiding disruptions, and improving the efficiency of biogas reactors (Wang et al., 2018). To enhance methane production, it is essential to investigate the structure and interactions between microbial communities and the operating conditions under which they function (Ziganshina et al., 2022; Kang et al., 2019).

This study examined the AD of PL at different solid concentrations (10 %, 30 %, and 60 %), incorporating co-digestion with food waste hydrolysate (FH) to optimise the C/N ratio, and introducing sulfur-oxidizing bacteria (SOB) with micro-aeration for in situ desulfurization and methane yield enhancement. The results provide a strong foundation for advancing SSAD of PL. Uniquely, this work integrates high-solids AD with FH co-digestion, SOB-mediated H_2S removal, and micro-aeration in a single system, coupled with microbial community profiling using high-throughput sequencing (454 pyrosequencing and Illumina). This approach achieves enhanced methane yield, maximum H_2S removal efficiency, and offers the first comprehensive insight into microbial diversity and its role in high-solids PL digestion for biofuel production.

2. Materials and methods

2.1. Feedstock procurement and sample preparation

Approximately 10 kg of PL was collected from a poultry farm from the outskirts of Hyderabad city in Telangana State, India. The litter was placed in a sealed bag and transported to the laboratory within a few hours of collection. The PL comprised of cage litter, which included bedding material such as straw, wood shavings, and sawdust, along with poultry droppings. The collected PL was ground to ensure homogeneity and stored in a refrigerator at 4 °C for subsequent biomethanation experiments. The raw PL was initially characterized for parameters such as TS and VS using standard methods as described in APHA. For this, 10 % TS containing PL slurry was prepared with deionized (DI) water, and was used for all subsequent characterization of raw PL. Prior to the experiments, PL slurry of 10 % TS, 30 % TS, and 60 % TS was prepared 12 h in advance, adjusted to meet the TS requirements of the study, and its pH was acclimatised by adding dilute hydrochloric acid from an initial value of 7.9 to an optimal 7 for AD.

The co-substrate used in the experiments was food waste hydrolysate (FH), which served as a carbon source to dilute the PL as well as to adjust pH. Food waste (FW) is rich in carbon (C), making FH an effective means of balancing the C/N ratio of PL, which is 8.9. Approximately 5 kg of coarsely ground FW was collected from a biogas plant operated in CSIR-IICT, which processes kitchen waste (cooked and uncooked food waste) under mesophilic (35 ± 2 °C) conditions. To prepare FH, 1 kg of FW was mixed with 1 L of distilled water to form a slurry, which was homogenized and passed through a mesh cloth to remove solids. The resultant filtrate was collected and used in co-digestion experiments to dilute PL.

2.1.1. Mixed microbial consortia and inoculum preparation

Mixed microbial consortia (MMC) were collected from an existing biogas plant based on an anaerobic gas lift reactor (AGR) operating at the Bowenpally vegetable market under mesophilic conditions (35 ± 2 °C) using vegetable waste as the substrate. The MMC was acclimatized with poultry PL for 7 days at mesophilic (35 ± 2°C) temperature to serve as the inoculum for the study. After acclimatization, the inoculum was characterized for pH and VS which were found to be 7.1 and 6 %, respectively. Additionally, MMC containing SOB was isolated from a previous study, with detailed isolation procedures available in Bhavya et al. (2024). MMC-SOB was used in a 1:1 ratio in reactors labelled St-PL (10 %, 30 %, 60 %) to assess its effectiveness in inhibiting H_2S production.

2.2. Experimental setup and operational procedures for poultry litter and food waste hydrolysate biomethanation

All SSAD batch experiments were conducted in 1 L glass reactors

Table 1
Details of solid state anaerobic batch reactors.

Rector Name	Composition (Substrate)	Inoculum in the reactors
PL-10 %	Poultry litter (100 g)	AGR (MMC) (400 mL)
PL-30 %	Poultry litter (200 g)	AGR (MMC) (400 mL)
PL-60 %	Poultry litter (400 g)	AGR (MMC) (400 mL)
PL + FH-10 %	Poultry litter (50 g) & Food waste hydrolysate (50 mL)	AGR (MMC) (400 mL)
PL + FH-30 %	Poultry litter (100 g) & Food waste hydrolysate (100 mL)	AGR (MMC) (400 mL)
PL + FH-60 %	Poultry litter (200 g) & Food waste hydrolysate (200 mL)	AGR (MMC) (400 mL)
St-PL-10 %	Poultry litter (100 g)	AGR (MMC) (200 mL) + SOB (MMC) (200 mL)
St-PL-30 %	Poultry litter (200 g)	AGR (MMC) (200 mL) + SOB (MMC) (200 mL)
St-PL-60 %	Poultry litter (400 g)	AGR (MMC) (200 mL) + SOB (MMC) (200 mL)
FH	Food waste hydrolysate (400 g)	AGR (MMC) (400 mL)

equipped with arrangements for inlet, outlet, and biogas collection ports. Biogas generated from the SSAD reactors was continuously measured using the water displacement method over the 30-day experimental period. A total of 10 SSAD reactors were operated using PL and FH + PL at varying solids concentrations (10 %, 30 % and 60 %) at mesophilic temperature. Experimental details are summarized in Table 1. The initial pH of each reactor was adjusted to 7 ± 0.2 and sparged with N_2 gas for 2 min to ensure anaerobic conditions. For reactors with stripping (St-PL at 10 %, 30 %, and 60 % solids concentrations), daily 15 min of micro-aeration at a flowrate of 0.005 L/min was introduced into the reactor headspace to maintain partial aerobic conditions for SOB activity. Rubber stoppers were used to seal the reactors tightly, with appropriate piping for gas collection and measurement. A substrate-to-inoculum (S/I) ratio of 1:1 was maintained across all reactors to balance microbial activity and substrate loading, minimizing lag phase and inhibition. All reactors were placed in an orbital shaker at 70 rpm under mesophilic temperature conditions (35 ± 2 °C). Each experiment was performed in triplicate for consistency, and average values were used for calculations. Reactor pH was monitored every alternate day, while biogas production and methane content were analysed every two days. Samples (~5 g) were collected from each reactor every 10 days and analysed for TS, VS, COD, VFA, and ammoniacal nitrogen. Methane yield was calculated based on the VS remaining in the reactor. A standalone reactor was operated with MMC to estimate the biogas generation potential of the inoculum. The cumulative biogas generated by the inoculum was subtracted from the biogas generated in each experimental reactor to determine the net biogas production from PL.

2.3. Analytical parameters and procedures

The raw feed stock (i.e., PL) was initially characterized for TS, VS and moisture content (MC), VFA, Alkalinity using standard APHA 2005 procedures (Baird et al., 2017). PL was also analysed for pH, alkalinity, chemical oxygen demand (COD), and ammoniacal nitrogen (A.N) by diluting the sample 10 times with DI water, as per APHA standard procedures (2005). Elemental analysis for carbon (C), hydrogen (H), nitrogen (N), and sulfur (S) was performed using a CHNS analyser (PerkinElmer, 2400 Series), with helium as the carrier gas and a thermal conductivity detector. Total volatile fatty acids (TVFAs) in the samples were determined using high-performance liquid chromatography (HPLC) (Shimadzu Model LC 10A VP) equipped with a UV detector (210 nm) and ODS column (250 mm × 4.6 mm ID). A buffer solution of 0.01 M H_2SO_4 in acetonitrile was used as the mobile phase at a flow rate of 0.6 mL/min, with a sample injection volume of 20 μL. Gas chromatography (GC) (Shimadzu GC-2010) was employed to analyse the biogas composition. The instrument was equipped with a thermal conductivity detector (TCD) and a Supelco Q Plot Biogas column (30 m × 0.53 mm). Nitrogen served as the carrier gas, with the column maintained at 38 °C, and detection and injection temperatures set at 250 °C and 200 °C, respectively. The injection volume was 1 mL. Heavy metals in the raw PL were quantified using inductively coupled plasma mass spectrometry (ICP-MS; model ICAP 6500 DUO). After the termination of the AD experiments, the final digestate was analysed to evaluate its quality and suitability as an organic fertiliser (bio-manure). Parameters such as nitrogen (N), potash oxide (K_2O), and phosphorus pentoxide (P_2O_5) were assessed in accordance with the Fertilizer Control Order (FCO) standards using ICP-MS. All analytical measurements were performed in triplicate to ensure accuracy, and the average values are reported. The chemicals used throughout the experiments were of analytical reagent grade.

2.4. Microbial community characterization using next generation sequencing

The raw PL collected from a local poultry farm, raw FH used for the experiments, and microbial sludge from the PL-30 %, PL + FH-30 %, and St + PL-30 % reactors three representative reactors with the highest performance in each category for sequencing, were collected on the 20th day when reactors reached steady- state methane production and stored at −20 °C in a 1:1 ethanol ratio to preserve samples for DNA isolation and subsequent characterisation, following the protocol outlined by Zhang et al. (2019a), Zhang et al. (2019b). The samples were gently thawed, and approximately 200 mg of each was used for DNA isolation, which was performed using the NucleoSpin® Soil Kit according to the manufacturer's instructions (Roopnarain et al., 2017). The purity of the isolated DNA was quantified using the NanoDrop spectrophotometer method (Alessi et al., 2020). Polymerase Chain Reaction (PCR) was conducted to amplify the DNA isolated from the reactor samples for metagenomic studies aimed at understanding the diversity of bacterial species present in the reactors. A dual indexing PCR sequencing strategy was used to analyse the amplified DNA fragments on the Illumina sequencing platform. Amplicons were purified using the QIAGEN Purification Kit, and their purity and quantity were assessed with a NanoDrop method. The PCR cycling conditions included: an initial cycle at 95 °C for 2 min, followed by 25 cycles of 95 °C for 15 s, 60 °C for 15 s, and 72 °C for 2 min, with a final extension at 72 °C for 10 min and a hold at 4 °C. Indexing primers targeting the V3-V4 variable region of the 16S rRNA bacterial gene (16S_F: 5′-AGAGTTTGATGMTGGCTCAG-3′; 16S_R: 5′-TTACCGCGGCMGCSGGCAC-3′) were used to generate amplicon libraries. Following indexing PCR, the constructed libraries were quantified using the Qubit fluorometer and the Agilent 2100 Bioanalyzer system (Model 2939A). Sequencing of the enriched libraries was performed at Biokart India using the Illumina HiSeq 2000 platform. The sequencing data obtained were analysed for alpha diversity (Chao1, PD whole tree, observed OTUs), which was calculated using R software. Beta diversity and taxonomic analysis were also performed. The summary of species diversity was examined using the QIIME platform, an advanced bioinformatics tool.

2.5. Statistical analysis

2.5.1. Kinetic study on the performance of solid-state anaerobic digestion reactors using three model equations

In this study, three kinetic models-the modified Gompertz model, the logistic model, and the transfer function model as shown in equations (Eq. 1, Eq. 2 and Eq. 3), were applied to fit experimental data on biogas production from PL. These models were used to evaluate the lag phase of the digestion process, predict the maximum biogas generation rate, and provide insights into the dynamics of biogas production.

Each kinetic model offered specific advantages. The modified Gompertz model, widely recognised as an effective empirical non-linear regression model, was specifically employed to simulate methane

Table 2
Characteristics of raw PL & FH.

Parameter	Unit	PL	FH	SD (PL)	SD (FH)
pH	–	7.9	5.2	7.90 ± 0.40	5.20 ± 0.30
Total Solids (TS)	%, wet basis	66	6.5	66.0 ± 3.3	6.50 ± 0.30
Volatile Solids (VS)	%, TS	26	4.9	26.0 ± 1.3	4.90 ± 0.20
Moisture Content (MC)	%, wet basis	34	93.5	34.0 ± 1.7	93.5 ± 4.7
VS/TS	–	0.39	0.75	0.39 ± 0.02	0.75 ± 0.04
Chemical oxygen demand (COD) as O_2	mg/L	1,40,000	80,000	140,000 ± 7,000	80,000 ± 4,000
Soluble COD (sCOD) as O_2	mg/L	80,000	25,000	80,000 ± 4,000	25,000 ± 1,200
Alkalinity as $CaCO_3$	mg/L	4,000	4,500	4,000 ± 200	4,500 ± 225
Volatile fatty acids (VFA) as CH_3COOH	mg/L	5,681	5,000	5,681 ± 284	5,000 ± 250
VFA/Alkalinity	–	1.4	1.11	1.40 ± 0.07	1.11 ± 0.06
Total Nitrogen (Kjeldahl, TKN) as N	mg/L	1,815	1,500	1,815 ± 91	1,500 ± 75
Ammoniacal nitrogen (NH_3-N) as N	mg/L	1,500	500	1,500 ± 75	500 ± 25
Sulfate (SO_4^{2-})	mg/L	5,000	450	5,000 ± 250	450 ± 23
Nitrite (NO_2^-) as N	mg/L	800	45	800 ± 40	45 ± 2
Nitrate (NO_3^-) as N	mg/L	5,000	90	5,000 ± 250	90 ± 4
Carbon (C)	%, dry basis	18.76	35	18.76 ± 0.94	35.0 ± 1.75
Hydrogen (H)	%, dry basis	3.129	5	3.129 ± 0.160	5.00 ± 0.25
Nitrogen (N)	%, dry basis	2.09	1.2	2.09 ± 0.10	1.20 ± 0.06
Sulfur (S)	%, dry basis	1.42	0.1	1.42 ± 0.07	0.10 ± 0.005
C/N	–	8.9	29.1	8.90 ± 0.45	29.1 ± 1.46
Cadmium (Cd)	mg/L	0.001	0.0001	0.001 ± 0.00005	0.0001 ± 0.000005
Chromium (Cr)	mg/L	0.084	0.001	0.084 ± 0.0045	0.001 ± 0.00005
Copper (Cu)	mg/L	0.033	0.05	0.033 ± 0.0017	0.050 ± 0.0025
Lead (Pb)	mg/L	0.006	0.001	0.006 ± 0.0003	0.001 ± 0.00005
Nickel (Ni)	mg/L	0.035	0.005	0.035 ± 0.0017	0.005 ± 0.0025
Zinc (Zn)	mg/L	0.134	0.01	0.134 ± 0.007	0.010 ± 0.0005

accumulation. Non-linear regression analysis was conducted using SPSS software (IBM SPSS) to determine key kinetic parameters, including the lag phase duration (λ), maximum biogas generation rate (Rm), rate constant (k), and predicted biogas yield. To assess and compare the accuracy and reliability of the models in predicting biogas production dynamics, the coefficient of determination (R^2) was calculated for each model. These analyses provided insights into the model performance and suitability for describing the PL digestion.

$$\text{Modified Gompertz model}: P = P_o.exp\left\{-\exp\left[\frac{R_m.e}{P_o}(\lambda - 1) + 1\right]\right\} \quad (1)$$

$$\text{Logistic model}: P = \frac{P_o}{1 + exp\left\{\frac{4.R_m.(\lambda - t)}{P_o} + 2\right\}} \quad (2)$$

$$\text{Transfer function model}: P = P_o.\left\{1 - exp\left[\frac{-R_m}{P_o}.(t - \lambda)\right]\right\} \quad (3)$$

where **'P'** is the biogas yield (L/g VS_{added}) with respect to time t (days); **'P_o'** is the maximum biogas potential of the substrate (L/g VS_{added}); **'k'** is the hydrolysis rate constant (1/day); **'t'** is the time (day); **'R_m'** is the maximum biogas production rate (L/g VS_{added}.d); **'λ'** is the lag phase time (days); **'e'** is Euler's function equal to e = 2.7183.

3. Results and discussion

3.1. Feedstock characterization and its significance in solid state anaerobic digestion

The characteristics of the substrate play a crucial role in determining its potential for biogas production. PL was found to be biodegradable with potential for biogas production. Numerous studies have identified PL as an excellent substrate, due to its high biodegradable content, which can be effectively harnessed for biofuel generation. A detailed summary of PL characteristics is provided in Table 2. The pH of PL was found to be 7.9, which is within the optimal range for AD. However, the VS/TS ratio (0.39) and C/N ratio (8.9) are relatively low, raising concerns about microbial activity and process stability. To address this, co-digestion with FH was introduced. The VFA/alkalinity ratio of PL was 1.4, indicating high VFA production. This necessitates careful pH monitoring to maintain the buffering capacity of the AD system. Stable mesophilic temperature and adequate retention time are critical for promoting methanogens to convert VFAs into methane effectively. The high ammoniacal nitrogen (1500 mg/L) and sulfate (5000 mg/L) content in PL present additional challenges. Ammonia inhibits methanogens, which are more sensitive to its effects compared to acidogenic and acetogenic bacteria. Similarly, high sulfate levels lead to H_2S production by SRB, adversely impacting both the AD process and biogas quality. To mitigate these issues, an SOB consortium isolated from different sulfur rich sources and enriched with aerobic culture media (ACM) was introduced for desulfurisation in the AD reactors (St-PL at 10 %, 30 %, and 60 % solids concentrations). The elemental composition of PL was determined to be 18.76 % carbon, 3.12 % hydrogen, 2.09 % nitrogen, and 1.42 % sulfur. Trace metal analysis showed cadmium, chromium, copper, lead, nickel, and zinc at 0.001 mg/L, 0.001 mg/L, 0.084 mg/L, 0.033 mg/L, 0.006 mg/L, and 0.134 mg/L, respectively. Additionally, the COD, alkalinity, and VFA concentrations were found to be 140,000 mg/L, 4000 mg/L as $CaCO_3$, and 5681 mg/L as CH_3COOH, respectively (Table 2). Metagenomic studies were conducted to investigate the impact of microbial consortia on biogas generation and to examine shifts in bacterial diversity at different solids concentrations.

3.2. Impact of substrate solid concentration and co-digestion on methane yield and biogas generation potential

The AD experiments were conducted with PL at TS concentration of 10 %, 30 % and 60 % in nine batch reactors along with one reactor containing food waste FH under mesophilic conditions (35 ± 2 °C). The reactors were designated as PL-10 %, PL-30 %, PL-60 %; PL + FH-10 %, PL + FH-30 %, PL + FH-60 %; St + PL-10 %, St + PL-30 %, St + PL-60 %, and FH. The daily biogas production and cumulative biogas yield over the 30-day experimental period are graphically represented in Fig. 1 (a) and (b). Peak biogas generation varied from 270 mL to 500 mL between days 14 and 16. The cumulative biogas and methane concentration were as follows: 5.9 L (47 % CH_4), 9.9 L (49 % CH_4), 5.39 L (45 % CH_4), 11.5 L (53 % CH_4), 12.4 L (57 % CH_4), 6.69 L (55 % CH_4), 5.4 L (61 % CH_4), 8.9

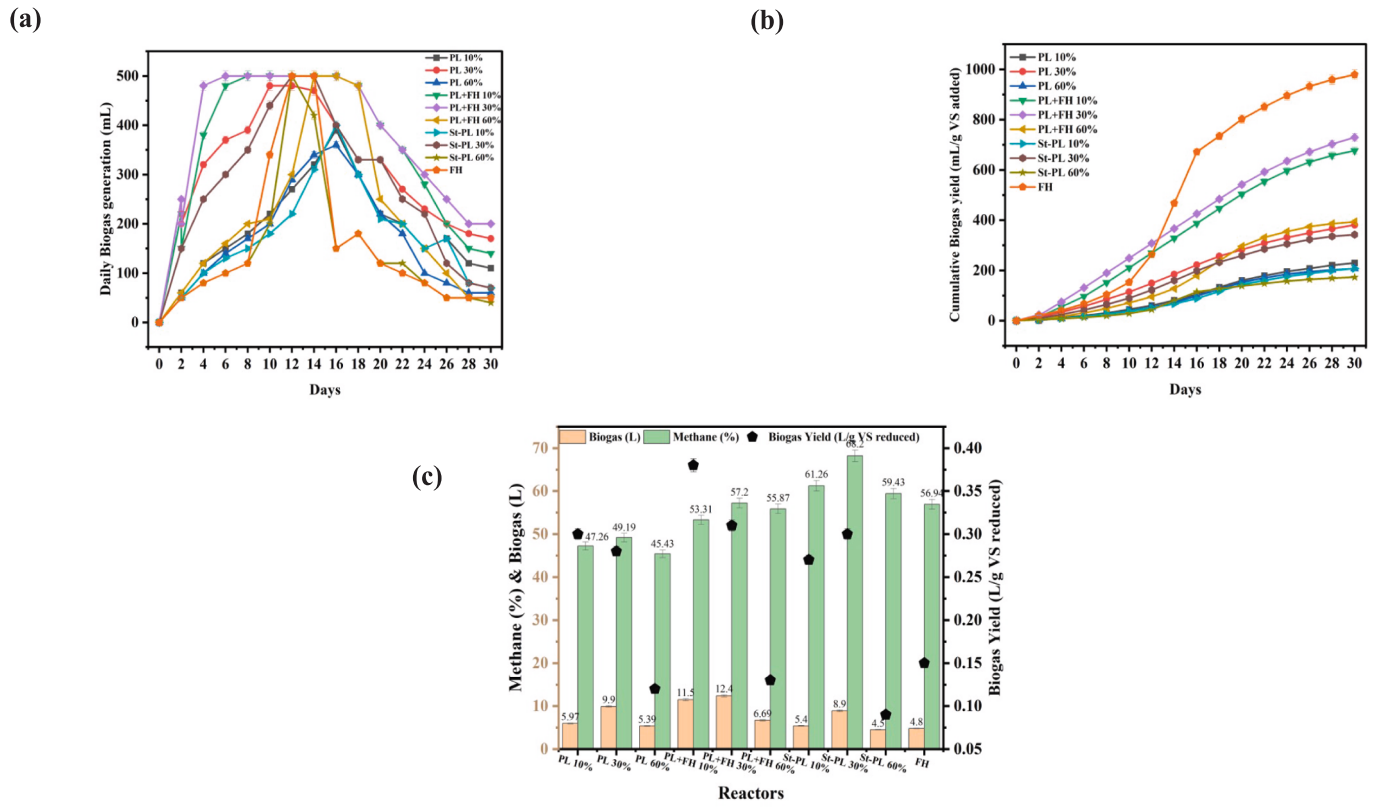


Fig. 1. (a). Daily biogas production, 1(b). Cumulative biogas production, 1(C) Biogas (L); Biogas yield & methane concentration.

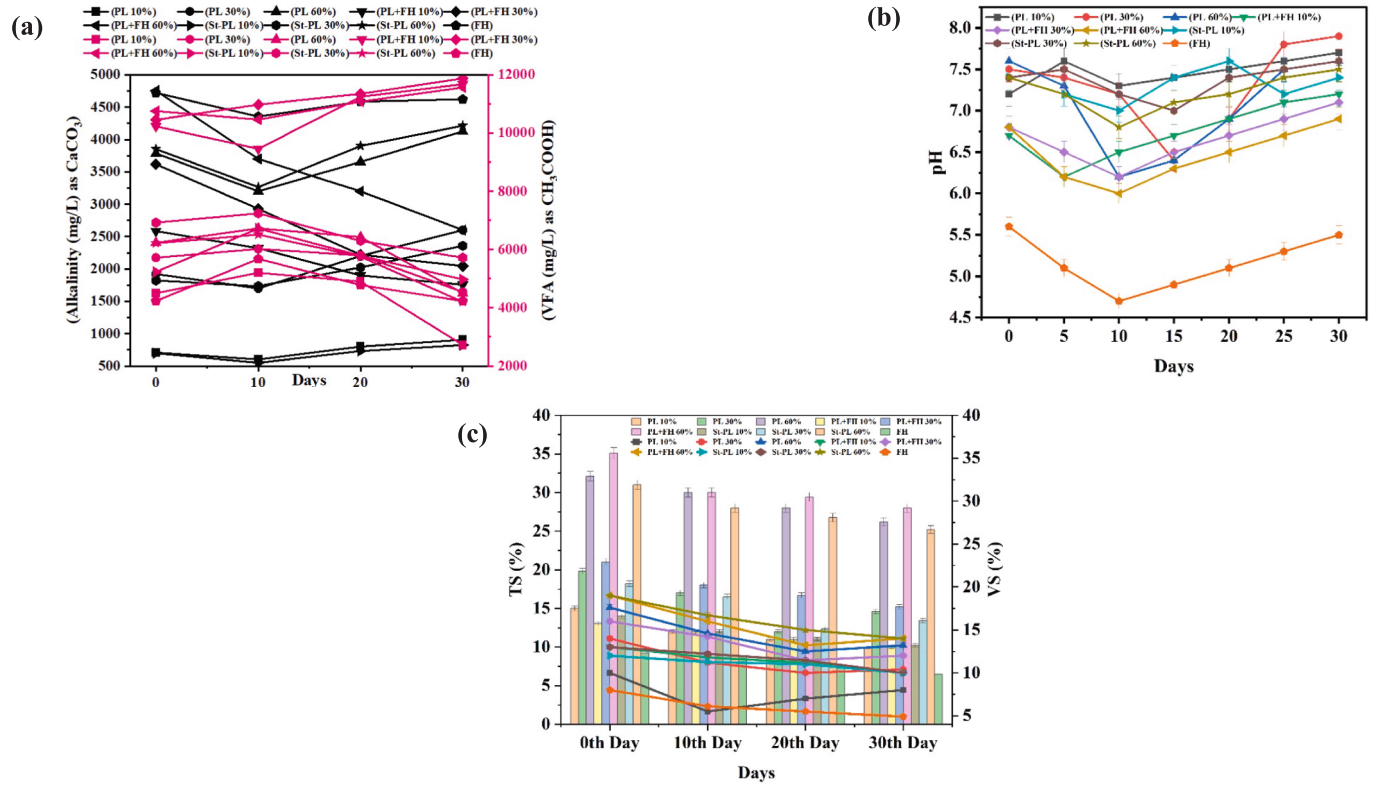


Fig. 2. (a) Alkalinity & VFA profile; (b) pH profile; (c) TS % and VS% profile.

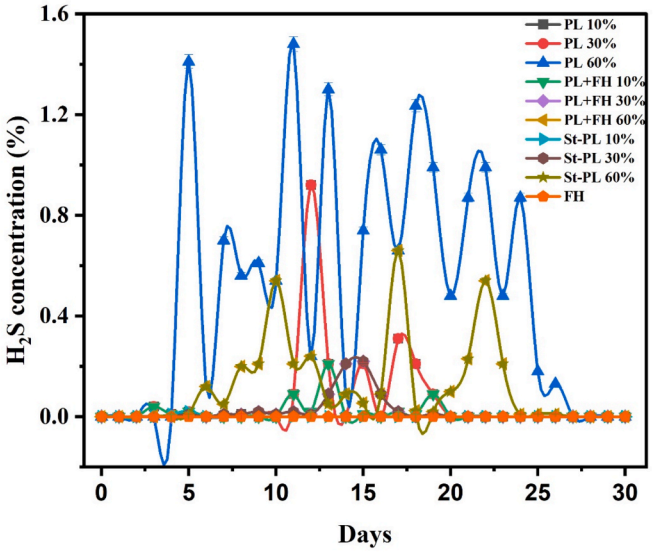


Fig. 3. H_2S concentration in biogas across reactors.

L (68 % CH_4), 4.5 L (59 % CH_4), and 4.8 L (56 % CH_4) for reactors PL-10 %, PL-30 %, PL-60 %; PL + FH-10 %, PL + FH-30 %, PL + FH-60 %; St + PL-10 %, St + PL-30 %, St + PL-60 %, and FH, respectively. Among all reactors, PL + FH-30 % recorded the highest cumulative biogas yield, followed by PL + FH-10 % and PL-30 %. Fig. 1(c) depicts the specific methane concentration and biogas yields in terms of liters per gram of volatile solids (L/g VS) reduced from each reactor. The results indicated that increasing the solids concentration in the reactor initially enhanced methane yield due to the greater availability of feedstock for microbial activity. However, at 60 % solids concentration, the digestion process was adversely affected, likely due to microbial inhibition from mass transfer limitations. The composition of PL at these solids concentration, combined with co-digestion, has limited studies available for direct comparison. However, comparable trends were observed in previous studies reported in literature. For instance, several researchers (Abbassi-Guendouz et al., 2012; Di Maria et al., 2014; Sun et al., 2016; Capson-Tojo et al., 2017) have observed that solids concentrations up to 25 % resulted in higher methane yield, whereas higher solids concentration were associated with reduced methane production. The experimental results from reactors with 30 % solids concentration, namely PL-30 %, PL + FH-30 %, and St + PL-30 %, demonstrated high performance in terms of both biogas generation and methane yield compared to other reactors.

The stability and operational mechanisms of the AD reactors were assessed by tracking key performance indicators such as alkalinity, VFA, TS, and VS, measured on the 10th, 20th, and 30th days. Additionally, pH was monitored at 2-day intervals. These results are illustrated in Fig. 2 (a-c). As shown in Fig. 2(a), the alkalinity levels in most reactors remained relatively stable. However, in the PL + FH 10 %, PL + FH 30 %, and PL + FH 60 % reactors, showed a measurable decline in alkalinity, consistent with the accumulation of VFAs during early-stage hydrolysis and acidogenesis processes occurring in these reactors.

This interpretation is further supported by the increased VFA levels in the PL + FH reactors, which followed trends similar to those seen in the other reactors. Fig. 2(b) shows the variation in pH across the reactors. The pH levels in the PL and St-PL reactors were stable, ranging from neutral to slightly basic. However, a gradual decrease in pH was observed in the PL-30 % reactor after 10 days. In contrast, the PL + FH reactors exhibited slightly acidic pH levels throughout the experiment, likely due to the accumulation of VFAs. The VFA profile, illustrated in Fig. 2(a), shows a notable increase in VFAs during the first 10 days, followed by a slight decline. The initial 10-day period was characterized by a reduction in pH, an increase in VFAs, and a decline in alkalinity, indicating the active presence of acidogenic and hydrolytic bacteria. The variation in the VFA was due to diversity of substrate fractions and biodegradability, persistence of VFAs after the methane plateau linked to residual hydrolysis and partial acid inhibition at higher solid load. Subsequently, the observed increase in pH, a slight reduction in VFAs, and a rise in alkalinity suggest the transition to methanogenesis as the dominant process within the reactors. Fig. 2(c) highlights the reduction in TS and VS levels across all reactors, except for the 60 % TS reactors. The reduced performance in the 60 % TS reactors can be attributed to the very high solids concentration, which likely impeded microbial activity and mass transfer efficiency.

3.3. The effect of micro-aeration and sulfur oxidizing bacteria addition on H_2S removal in biogas

The H_2S levels in the biogas obtained from reactors labelled as St-PL (10 %, 30 %, 60 %) following the implementation of micro-aeration and SOB are illustrated in Fig. 3. In other reactors, daily H_2S emission ranged from 0.2 % to 1.5 %, and these values were used as a baseline to determine the H_2S removal efficiency in the St-PL reactors. The desulfurization efficiency in the St-PL reactors ranged from 80 % to 92 %, as evident in Fig. 3. Although the H_2S concentration decreased significantly compared with the other reactors, complete removal of sulfide from the biogas was not achieved. The highest removal efficiency was observed in the St-PL-30 % reactor. H_2S is a metabolic byproduct of SRB, and the observed reduction in cumulative H_2S production in the St-PL reactors can be attributed to the inhibition of SRB activity. Previous studies have shown that low concentration of oxygen exerts toxic effects on SRB by generating reactive oxygen species and competing with sulfate as an electron acceptor. This interference inhibits dissimilatory sulfate reduction and may inactivate SRB (Brioukhanov & Netrusov, 2007). The maximum desulfurisation efficiency achieved in this study was 92 %, which is notably higher than the values reported by Zhou et al. (2007). Conversely, the activity of SOB led to the conversion of gaseous H_2S into sulfate or elemental sulfur in the digestate. As a result, the sulfate concentration in the digestate of St-PL reactors were consistently higher than those in other reactors, as will be discussed in Section 3.8. The reduction in H_2S also increased the availability of methanogenic substrates, likely enhancing methanogenic activity under micro-aerated conditions. Notably, the relative abundance of hydrogenotrophic methanogens, particularly *Methanocellus*, increased significantly in the St-PL reactors, as elaborated in the subsequent section. The interplay between the activities of SRB and methanogens in utilising electron donors has been previously reported by Damianovic and Foresti (2009). The high sulfate concentrations observed in the St-PL reactors may also reflect sustained methanogenic activity, contributing to higher methane yields. In addition to achieving significant H_2S removal, the introduction of micro-aeration and SOB consortia in the St-PL reactors resulted in a notable enhancement in methane production. This dual benefit underscores the effectiveness of this strategy in improving biogas quality and reactor performance.

Several studies have highlighted the significant impact of introducing small amounts of air or oxygen into AD processes on performance. However, the level of enhancement observed in this study exceeded those reported previously, particularly in the St-PL 30 % reactor, which achieved the highest methane concentration of 68.2 %, compared to other reactors in this study, where methane concentrations ranged from 45 % to 61 %. For instance, Tsapekos et al. (2017) demonstrated a 7.2 % increase in methane yield when 5 mL O_2/g VS was introduced into an AD system processing wheat straw over three days. Similarly, Lim & Wang (2013) reported a 21 % improvement in methane production from brown water and food waste under micro-aeration (37.5 mL O_2/L reactor/day for four days) compared with strictly anaerobic conditions. These findings, along with the results of the present study, highlight the critical role of substrate characteristics in determining the effectiveness of micro-aeration strategies. PL, which is

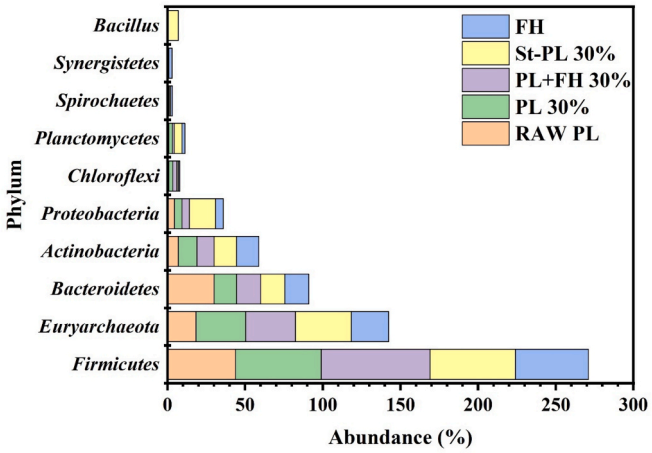


Fig. 4. Illustration of abundant Microbial diversity at phylum level in reactors.

rich in ammonia (as shown in Table 2) and sulfur-containing amino acids, is particularly susceptible to the synergistic inhibitory effects of sulfide and ammonia during AD, as noted by Sürmeli et al. (2019). Additionally, the degradation of substrates like PL leads to high levels of sulfur compounds, ammonia (NH_3), and VFAs, creating conditions that promote H_2S production and inhibit methanogenesis. The outcome and benefits of micro-aeration strategies are highly substrate-specific, with more pronounced effects observed in systems processing substrates like PL. This emphasizes the necessity of tailoring micro-aeration approaches to the unique characteristics of feedstock.

3.4. *Effect of microbial consortia on the methanogenesis process: Microbial diversity in reactors*

Metagenomic studies were conducted to analyse the microbial communities within the reactors. Five samples-PL-30 %, PL + FH 30 %, and St-PL 30 %, which demonstrated the highest biogas production and H_2S removal efficiency-were selected for analysis. Additionally, raw PL and raw FH samples were included to compare microbial shifts between the raw substrates and reactor samples after digestion. The shifts in microbial community distribution at the phylum level are illustrated in Fig. 4. As shown in Fig. 4, the relative abundance of bacterial communities at the phylum level reveals substantial differences between the raw samples and reactor sludge samples. In all the AD reactors, *Firmicutes* emerged as the most abundant bacterial phylum, accounting for 43–70 % of total reads. This was followed by *Bacteroidetes* (30–36 %) and *Actinobacteria* (7–14 %). Other notable bacterial phyla included *Chloroflexi* (1–2 %), *Spirochaetes* (1–2 %), and *Synergistetes* (0.4–1 %), which are frequently reported in AD processes (Lee et al., 2016; Yin et al., 2018). *Firmicutes* play a key role in AD by breaking down complex organic substrates into intermediates such as H_2, acetic acid, CO_2, and butyric acid, while *Bacteroidetes* are active during hydrolysis and acidogenesis, degrading lipids and cellulose compounds, which enhances substrate availability and overall biogas production (Zhang et al., 2019; Westerholm & Schnürer, 2019). Proteobacteria, another common hydrolytic bacterial group, utilise glucose, acetate, and propionate, and share metabolic pathways with *Firmicutes* during the acetogenesis phase. In addition to bacterial communities, archaeal communities play a significant role in methane production, with their abundance varying across different stages of the AD process.

The microbial diversity results differed significantly in the St-PL 30 % reactor due to the introduction of micro-aeration and the SOB consortia. In this reactor, the relative abundance of *Firmicutes*, particularly the order *Clostridiales*, increased. Similarly, members of the Proteobacteria phylum showed a 32 % rise in relative abundance under the same conditions. Previous studies have noted that many SRB and SOB are classified under *Proteobacteria*, and the SOB consortia used in this reactor contained approximately 80 % *Proteobacteria* (Pokorna & Zabranska, 2015; Bhavya et al., 2024).

The enhanced methane yield, faster hydrolysis kinetics, and maximum H_2S suppression observed in the St-PL 30 % reactor suggest that differences in bacterial richness and diversity likely existed, even in the absence of significant variations in phyla proportions, as shown in Fig. 4. Additionally, the activity of bacterial populations may have shifted. For instance, under oxidative stress, the growth rate of acetogens has been reported to decrease significantly without impairing their acetate synthesis activity (Brioukhanov & Netrusov, 2007). The St-PL 30 % reactor demonstrated the highest biogas production and H_2S suppression due to SOB activity. Micro-aeration likely facilitated an anaerobic micro-environment, where methanogens were shielded from oxidative stress primarily because of oxygen uptake by facultative bacteria within these phyla. These bacteria consumed oxygen, preventing it from directly harming methanogens and maintaining the local anaerobic conditions essential for methanogenic activity and biogas yield. This protective mechanism, critical to the success of the micro aeration strategy, relies on efficient oxygen utilisation by facultative bacteria, which create a “buffer zone” to ensure uninterrupted AD (Wanga et al., 1999). In this study, different SRB, including *Desulfovibrio*, *Desulfobulbus (Proteobacteria)*, and *Desulfitibacter (Firmicutes)*, were found in the digesters at relatively low abundances (<5%). Under micro-aerated conditions, their abundance decreased by approximately half. These genera, commonly found in sulfate reduction reactors, are classified as incomplete organic oxidisers. *Desulfobulbus* and *Desulfovibrio* species are known to utilise ethanol, lactate, and hydrogen as electron donors, with sulfate serving as the electron acceptor, producing acetate as a byproduct (Liu et al., 2018). The decline in hydrogen-utilising SRB may have been offset by an increase in hydrogen-utilising methanogens. Conversely, several SOB, including *Thioalkalimicrobium*, *Alishewanella*, *Bacillus (Firmicutes)*, and *Pseudomonas (Proteobacteria)*, were detected in the St-PL 30 % reactor. These SOB likely contributed to the enhanced H_2S removal efficiency by metabolising H_2S and converting it into other sulfur compounds, such as sulfate or elemental sulfur, depending on the environmental conditions and metabolic pathways. Further details on the digestate composition are discussed in the subsequent section.

3.5. *Comparison of archaeal community structures*

Methanogens are the sole archaeal group responsible for converting organic matter into methane and carbon dioxide, a process carried out in symbiosis with bacteria. The presence and abundance of specific methanogenic archaea in AD systems directly influence the rate and efficiency of methanogenesis. The most abundant archaeal communities identified in the reactors were *Methanosaeta* and *Methanobacterium*, both belonging to the phylum *Euryarchaeota*. Methanomicrobia is considered key methanogens in AD system (Kumar Awasthi et al., 2021). The phylum *Euryarchaeota* utilises intermediate products such as acetate, H_2, and CO_2 as substrates to produce methane through various methanogenic pathways. Metagenomic analysis of the reactor samples provided insights into the dominant methanogenic species and community shifts. The results revealed that *Euryarchaeota* was the predominant archaeal group in all reactors. Additionally, the presence of *Crearchaeota* and *Parvarchaeota*, both parts of the kingdom Archaea, was observed. Taxonomic analysis showed that *Euryarchaeota* accounted for 57 % of all archaeal organisms in the control reactor, increasing to 86 %, 84 %, and 80 % in reactors St-PL 30 %, PL + FH 30 %, and PL 30 %, respectively. The biogas production data correlated with the highest abundance of *Euryarchaeota* in reactor St-PL 30 %, followed by PL + FH 30 % and PL 30 %.

In raw PL and FH samples, *Methanosaeta* was the most dominant genus within *Euryarchaeota*, comprising 29–32 % of the archaeal community, followed by *Methanoculleus* (2–4 %), *Methanosarcina* (3–4 %), and *Methanocorpusculum* (2–4 %). Other methanogens, including

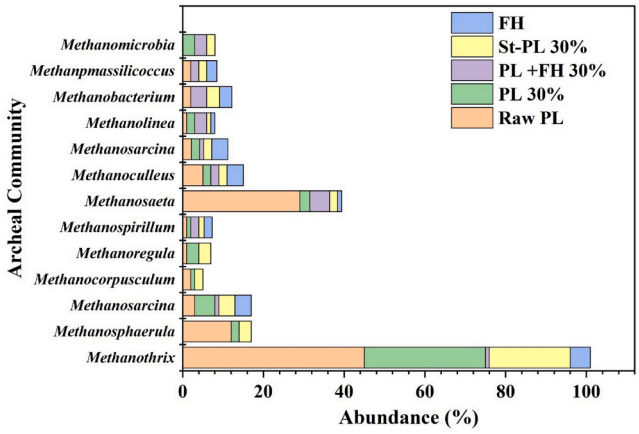


Fig. 5. Representation of Archaeal community abundance in reactors.

Methanobacterium, *Methanolinea,* and *Methanospirillum*, were also detected. Co-digestion and the introduction of SOB consortia to the reactors caused notable community shifts. The relative abundance of *Methanosaeta* significantly decreased across reactors, dropping to 2.5 % in PL 30 %, 2.4 % in PL + FH 30 %, and 4.4 % in St-PL 30 %, as shown in Fig. 5. *Methanosarcina* dominated all reactors except PL + FH 30 %, where *Methanobacterium*, known for its hydrogenotrophic methanogenesis, was the predominant genus (Li et al., 2020). In the PL + FH 30 % reactor, *Methanobacterium* and *Methanolinea* were abundant, while *Methanomassiliicoccus* was detected in all reactors.

3.6. Kinetic potential parameters of anaerobic digestion

The kinetic analysis of AD of PL was performed by using three kinetic models: the modified Gompertz, logistic, and transfer function models. These models were utilized to estimate the hydrolysis rate constant, lag phase, maximum methane production rate, and maximum methane yield under mesophilic conditions, as summarised in Table 3. Experimental biogas production from the reactors was compared with biogas yields predicted by these models. As shown in Fig. 6, the modified Gompertz, logistic, and transfer function models exhibited excellent agreement with the experimental data, with R^2 values exceeding 0.96. Figs. 6(a) to 6(j) demonstrate the overlap between experimental and predicted values for all reactors, further corroborating the data presented in Table 3.

The lag phase for all reactors was calculated, with the longest lag phase observed in the PL + FH 60 % reactor across all models. This extended lag phase can be attributed to the high solid's concentration of the substrate, which likely inhibited microbial activity. Among the three models, the modified Gompertz model provided the best fit for the experimental biogas yields (R2 = 0.99), with deviations remaining under 30 %, except for the FH reactor. While the logistic and transfer function models also performed well, the modified Gompertz model demonstrated superior accuracy in predicting biogas yields and lag phases.

3.7. Nutrient analysis of bio digestate of reactors and sulfate in the micro-aerated reactor (St-PL 30 %)

The final digestate from each reactor-Raw PL, PL-30 %, PL + FH 30 %, and St-PL 30 %-was analysed for nutrient content at the end of the experiments. The digestate demonstrated potential as biomanure based on NPK content, improving soil nutrient levels and fertility

Table 3
Summary of kinetic potential of AD reactors.

Kinetic Model	Parameter		Units	PL-10 %	PL-30 %	PL-60 %	PL + FW-10 %	PL + FW-30 %	PL + FW-60 %	St-PL-10 %	St-PL-30 %	St-PL-60 %	FWH
Modified Gompertz model	R^2			0.99	1.00	0.99	1.00	0.99	0.99	0.99	0.99	0.99	0.99
	λ		**d**	**11.67**	**3.55**	**7.33**	**3.68**	**2.65**	**7.64**	**7.21**	**4.78**	**7.69**	**7.98**
	R_m		$mL_N/g\ VS_{added}.d$	6.60	17.87	11.93	32.10	32.56	23.27	10.82	17.69	12.80	78.70
	Biogas yield	**Pred**	$mL_N/g\ VS_{added}$	233.09	379.86	212.65	680.77	727.72	407.26	211.64	347.35	171.97	968.38
		Exp	$mL_N/g\ VS_{added}$	229.62	380.77	207.31	676.47	729.41	393.53	207.69	342.31	173.08	979.59
		Diff	%	1.51	0.23	2.57	0.63	0.23	3.48	1.90	1.47	0.64	1.14
Logistic model	R^2			0.99	0.99	1.00	0.99	0.99	0.99	0.99	0.99	0.99	0.99
	λ		**d**	**7.85**	**4.58**	**8.45**	**4.70**	**3.61**	**8.81**	**8.48**	**5.84**	**8.25**	**8.43**
	R_m		$mL_N/g\ VS_{added}.d$	13.06	19.22	13.24	34.42	34.55	26.06	12.18	19.22	13.43	81.44
	Biogas yield	**Pred**	$mL_N/g\ VS_{added}$	227.26	371.68	207.12	667.22	714.85	396.31	206.17	339.02	167.42	944.1
		Exp	$mL_N/g\ VS_{added}$	229.62	380.77	207.31	676.47	729.41	393.53	207.69	342.31	173.08	979.59
		Diff	%	1.02	2.38	0.09	1.36	1.99	0.05	0.73	0.96	3.27	3.62
Transfer function model	R^2			0.98	0.99	0.97	0.94	0.99	0.96	0.97	0.98	0.95	0.94
	λ		**d**	**3.63**	**2.13**	**3.86**	**2.31**	**1.80**	**3.97**	**4.00**	**2.74**	**3.78**	**3.86**
	R_m		$mL_N/g\ VS_{added}.d$	9.07	17.35	8.53	31.36	3426	16.35	8.28	15.76	8.61	51.17
	Biogas yield	**Pred**	$mL_N/g\ VS_{added}$	239.26	394.34	222.95	706	749.84	425.59	215.31	363.62	187.79	1065.36
		Exp	$mL_N/g\ VS_{added}$	229.62	380.77	207.31	676.47	729.41	393.53	207.69	342.31	173.08	979.59
		Diff	%	4.19	3.56	7.54	4.36	2.80	8.14	3.66	6.22	8.49	8.75

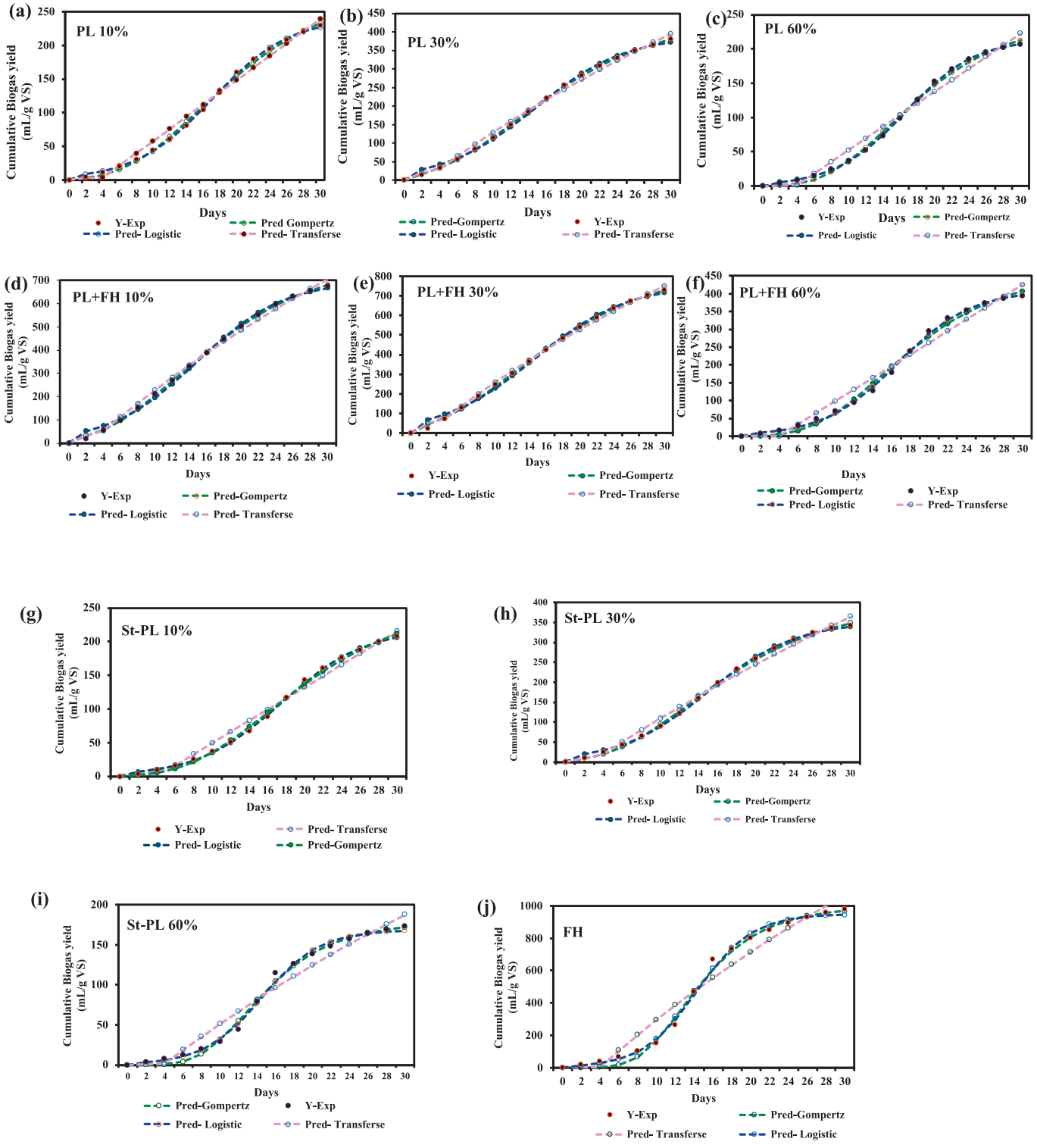


Fig. 6. (a-j). Plots illustrating the experimental biogas generation obtained from the AD of PL to that of predicted values using 3 models.

(Thiyageshwari et al., 2018). Nutrient analysis of the digestate, specifically for nitrogen (N), phosphorus pentoxide (P_2O_5), and potassium oxide (K_2O), revealed concentrations ranging from 0.4–0.55 % N, 0.6–0.8 % P_2O_5, and 0.9–1.2 % K_2O across all reactors. According to the Fertilizer Control Order (FCO) guidelines issued by the Ministry of Agriculture, India), the minimum nutrient requirements for biomanure are 0.5 % for each of N, P_2O_5, and K_2O, confirming that the digestate meets these standards. In addition, the digestate from the St-PL 30 % reactor was subjected to sulfur analysis to evaluate how the microbial community converted sulfur, leading to significantly suppressed H_2S concentration in the biogas. Sulfur analysis revealed traces of elemental sulfur (~45 %) and high sulfate concentrations (800 mg/L) compared to other digestate samples, as determined by ED-XRF analysis. These findings suggest that sulfur conversion processes within the reactor were effective; however, more rigorous molecular studies and continuous monitoring of the AD system's micro-environment will be necessary to precisely trace sulfur transformations within biodigesters.

Heavy metal analysis indicated that concentrations of lead, nickel,

cadmium, arsenic, chromium, zinc, and mercury were below detection limits, demonstrating that the digestate is safe for agricultural use. Elevated levels of these metals, if present, could adversely affect plant growth and pose risks to human health. Beyond its use as biomanure, the digestate holds potential for other applications, such as bioethanol production or microalgae cultivation (Gao & Li, 2011). The study also observed a reduction in digestate production at higher solids concentrations due to lower water usage, which facilitated easier transport and reduced energy requirements for reactor maintenance. These findings highlight that AD at high solids concentrations offers several advantages, including improved methane yield, increased biogas generation, and more efficient biodigester management. To optimise the utilisation of PL, a cost-effective, environmentally sustainable, and labour-efficient method should be adopted. Considering the environmental challenges associated with odour, pathogen contamination, and greenhouse gas emissions, AD of PL represents a beneficial approach for renewable energy production and sustainable waste management.

4. Conclusions

This study demonstrated that SSAD of poultry litter at high solids concentrations (up to 30 % TS) correlates positively with biogas yield, achieving 0.31 L/gVS$_{reduced}$ (PL + FH 30 %), 0.28 L/gVS$_{reduced}$ (PL-30 %), 0.3 L/gVS$_{reduced}$ (St-PL 30 %). The improved C/N ratio from 8.9 (PL) to 29.2 (PL + FH), contributed to the overall increase in biogas yield. The integration of SOB and micro-aeration effectively reduced H_2S levels in the biogas, enhancing its suitability as a biofuel. Metagenomic analysis revealed that microbial diversity and composition were influenced by solid concentrations and bioreactor parameters, highlighting their critical role in biogas generation efficiency. Additionally, the nutrient-rich digestate offers potential as a sustainable fertilizer alternative. Overall, the findings highlight the dual benefits of SSAD of poultry litter for efficient waste management and renewable energy production, offering a viable and sustainable approach to address environmental challenges while contributing to the circular bioeconomy.

CRediT authorship contribution statement

Bhavya Karumanchi: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sameena Begum:** Writing – review & editing, Visualization, Validation, Methodology, Data curation, Conceptualization. **Gangagni Rao Anupoju:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Biplob Kumar Pramanik:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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